Frank Qiu

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Research Interests

My research focuses on networking systems and machine learning systems. I enjoy taking a bottom-up approach, co-designing programmable network hardware with ML systems to improve performance across layers — from data-plane hardware and protocols up to collective communication and distributed training.

Education

Purdue University **September 2024 – Expected June 2029**
Ph.D. in Electrical and Computer Engineering, advised by Prof. Xiaoqi Chen *West Lafayette, Indiana, US*

Shanghai Jiao Tong University **September 2020 – June 2024**
B.E. in Computer Science and Technology (Zhiyuan Honors Program) *Shanghai, China*
ACM Class, advised by Prof. Yong Yu

Current Research

Trimmable Transport for Distributed ML Training **Manuscript in Preparation**

- Co-designed a trimmable transport and gradient codec to reduce collective completion time by avoiding retransmission delays: switches trim packets under congestion, while the GPU codec distributes gradient information across packets to enable graceful reconstruction from retained prefixes.
- Built a full-stack prototype on a physical 400 Gbps, 8×H100 testbed with a Tofino2 switch, spanning RoCEv2 via ibverbs, a BlueField-3 DPU dual receive path (ConnectX-7 & DPA), CUDA kernels with GPUDirect RDMA, and a PyTorch DDP hook.
- In scale-out htsim simulations with congestion from background jobs, kept communication slowdown within 2%, while RoCE, DCTCP, and EQDS degraded significantly.
- Pre-trained and fine-tuned diverse models, including LLM, MoE, VLM, and JEPA models; the pipelined, high-throughput implementation matches the end-to-end iteration time of NCCL and UCCL while staying robust to congestion.

Experience

Distributed Systems Laboratory, University of Pennsylvania **August 2023 – December 2023**
Research Assistant, advised by Prof. Vincent Liu *Philadelphia, Pennsylvania, US*

- Investigated co-design of the FPGA-based SmartNIC and the host networking stack.
- Leveraged idle PCIe transactional bandwidth to mitigate host congestion.

BCMI Laboratory, Shanghai Jiao Tong University **July 2022 – July 2023**
Research Assistant, advised by Prof. Li Niu *Shanghai, China*

- Proposed a reflectance-guided deep network for image harmonization.
- Generated diverse plausible foreground reflectances to enable multiple harmonization results.

Zhiyuan College, Shanghai Jiao Tong University **September 2021 – June 2023**
Teaching Assistant (Programming, Computer Architecture, Operating Systems, etc.) *Shanghai, China*

Skills

Engineering: Programmable network hardware (Intel Tofino2, NVIDIA BlueField-3 DOCA & DPA); host networking (RDMA ibverbs/RoCEv2, DPDK); GPU systems (CUDA, GPUDirect RDMA, GPUNetIO, NCCL, UCCL); distributed training (PyTorch); network and ML-training simulators (htsim, SimAI); end-to-end testbed bring-up and performance measurement.

Programming: C/C++, Python, Rust, Java, Verilog, etc.; algorithms and data structures.
Won First Prize in the *National Olympiad in Informatics in Provinces* (NOIP, Senior Division, Shanghai) in 2017 and 2018.